An Improved Fast Fractional Pel Motion Estimation Algorithm Based on H.264

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Abstract: **This paper presents an improved fast fractional pel motion estimation algorithm for H.264. First, to skip inefficient fractional pel search, smooth macro-block prediction algorithm is proposed. Then, to further reduce the computation load of H.264 fractional pel motion estimation, an extended diamond pattern search algorithm based on predicted motion vector is proposed. The simulation result shows that compared with the FFPS (Full Fractional Pel Search) algorithm, the proposed algorithm can save about 39.5% computation load of the whole motion estimation module only with PSNR degradation less than 0.189dB.**
Index Terms—**H.264/AVC, fraction pel motion estimation, fast search algorithm.**

I. INTRODUCTION

H.264/AVC is the newest video compression standard jointly developed by the ITU-T VCEG and the ISO/IEC MPEG. Similar to previous video standards such as H.261, H.263, MPEG-2, H.264/AVC is also based on hybrid coding framework. In H.264 video coding system, motion estimation (ME) can reduce the temporal redundancy among successive frames, the efficiency of which is highly related with coding speed, compression rate and image quality of decoded video. H.264 motion estimation module introduces many coding tools, including multiple reference frames, multiple block sizes, and quarter pel accuracy. Therefore, performance of H.264 codec exceeds all the previous codec. H.264/AVC baseline profile allows an average bit rate saving of about 40% compared with H.263 baseline with the same image quality [1]. The high performance of H.264 is achieved at the cost of the increasing of computational complexity, which is 4-5 times compared with H.263. Motion estimation module is the most computationally complex process that takes up 50-90% of the entire system [2].

In H.264 codec, motion estimation module is conducted in two parts: integer pel ME, and fractional pel ME. Fractional pel ME searches for fractional pel motion vector (MV) around integer pel MV, which can greatly improve the performance of codec on video quality and compression rate. Experiments show that fractional pel ME improves compression rate by 48%, and PSNR by 1-3dB on average than that of integer pel ME. However, because of extra computation of interpolation and fractional pel MV search, fractional pel ME imposes great computational load on the whole ME module.

Integer pel ME has been a hot issue ever since the emergence of H.264 standard. H.264 software JM currently supports two Fast Motion Estimation (FME) algorithms: UMHexagonS [3] and EPZS [4]. Compared with Full Search (FS) strategy, FME brings a great portion of ME time reduction (up to more than 90%), and total JM coding time is reduced by 60% on average. H.264 adopts variable block size motion compensation prediction (MCP) with seven prediction blocks modes. If classic full fractional pel search of quarter pel accuracy is adopted, each macro-block has to search 112 points (16 points of each mode). Since the complexity of integer pel ME has been greatly reduced by FME, improvement of fractional pel ME algorithm becomes crucial to the optimization of ME module.

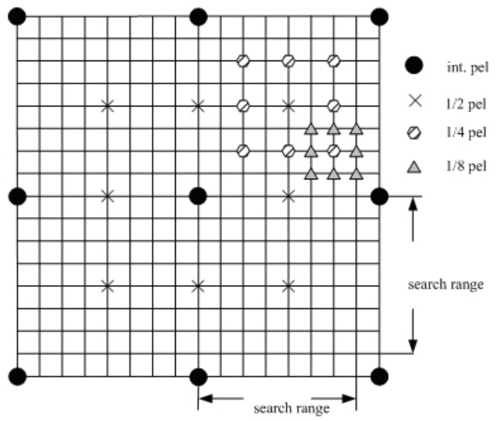
The rest of the paper is organized as follows. In section II, fractional pel ME algorithms adopted by H.264 are introduced. In section III, properties of H.264 fractional pel ME are analyzed. In section IV, we propose an improved fast fractional pel motion estimation algorithm, which skips inefficient fractional pel search according to video content, and adopts an extended diamond pattern search algorithm. The simulation results are shown in Section V and conclusion is outlined in Section VI.

II. H.264 FRACTIONAL PEL ME ALGORITHM

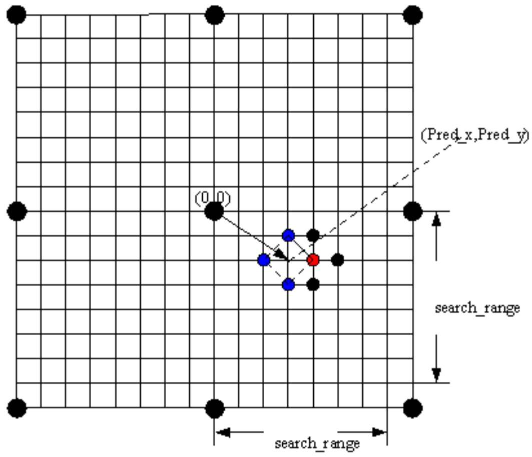
H.264/AVC adopts two fractional pel ME algorithm: Full Fractional Pel Search (FFPS) and Center Based Fractional Pel Search (CBFPS) [3].

Fig.1(a) illustrates the implementation of FFPS algorithm. The FFPS algorithm is described in the following steps: first, check the eight 1/2-pel positions around the best integer pel position in order to find the best 1/2-pel MV; second, check the eight 1/4-pel positions around the best 1/2-pel position in order to find the best 1/4-pel MV; third, select the MV and block-size mode, which produces the lowest rate-distortion cost. 16 fractional pel points need to be checked. The computation load of fractional pel motion estimation is becoming the bottleneck of improving the speed of motion estimation.

Fig.1(b) illustrates the implementation of CBFPS algorithm. The whole algorithm is described in the following steps: step 1, predict the MV of the current block by the median value of MVs of the adjacent blocks (the predicted motion vector is (Pred_x, Pred_y)); step 2, costs of the original search center (0, 0) and (Pred_x, Pred_y) are compared. The point with the minimum matching error is chosen as the search center; step 3, if the minimum block distortion (MBD) point is located at the center, go to step 4; otherwise choose the MBD point in this step as the center of next search, then iterate this step; step 4, choose the MBD point as the MV. CBFPS can save 20% computation overhead compared with FFPS with almost the same image quality.



(a) Implementation of FFPS algorithm



(b) Implementation of CBFPS algorithm

Fig.1. H.264 fractional pel ME algorithm

III. Analysis of Properties of H.264 Fractional Pel ME

A. Fractional Pel MV Prediction

The error surface of integer pel ME is not unmonotonic due to large search window and complexity of video content. So integer pel ME is prone to trap into a local minimum. Contrarily, since the sub-pels are generated from interpolation of integer pels, and the correlation inside fractional pel search window is much higher than that of integer pel search window. The matching error decreases monotonically as the search point moves closer to the global optimum [3]. Therefore, many fast fractional pel ME algorithms adopted fractional pel predicted MV (FMVP) as the original search point. If we can predict the original search point precisely, fractional pel MV search can be terminated earlier.

The FMVP of the current block can be determined by the median predictor obtained from neighbor blocks (top, top-right, left of the current blocks). FMVP contains the information of the predicted integer and fractional pel MV. We could extract the predicted fractional pel motion vector by using this formula:

$$frac\_pred\_mv = (pred\_mv - mv)\%\beta \qquad (1)$$

Where mv is the integer pel motion vector, and here mv is also in fractional pel unit, % is the mode operation, β=4 in 1/4-pel case and β=8 in 1/8-pel case.

TABLE I

Matching of FMVP & Best Fractional Pel MV

Sequence	Match	[-1, 1]	[-2, 2]
Tempete	82%	97%	98%
Silent	82%	96%	98%
Container	92%	99%	99%
Bus	46%	85%	94%
Football	31%	72%	92%

Table I shows the matching degree of FMVP and best fractional pel MV. A probability of FMVP being the best MV is larger than 82% in sequences of less detail and complex movement. Though matching degree of sequences of high detail and complex movement is lower, the probability of FMVP being the best MV is larger than 92% in [-2, 2]. Therefore, we select FMVP as the initial search center for FMV.

B. Partial Inefficiency of H.264 Fractional Pel ME

Fractional pel motion estimation can improve image quality of decoded video and compression rate, which, however, requires large computation. If the minimum sum of absolute difference (MSAD) obtained from fractional pel MV search is larger than that of integer pel MV search, integer pel MV is selected as the final MV, hence fractional pel MV search is ineffective. Six sequences (Tempete, Container, Silent are of less detail and complex movement; Bus, Stefan are of medium detail and complex movement; Football is of high detail and complex movement) are selected for the experiment, and ratio of integer pel MV in the final MVs is listed in Table II.

Sequences of less detail and complex movement, such as Tempete, Silent, Container, have over 80% of final MVs on integer pel position. Actually, for the smooth region of a image, contribution of fractional pel ME to coding efficiency is less prominent, and can be regarded as inefficient. If we can predict smooth regions, and skip ineffective fractional pel MV search, substantial unnecessary computation can be saved.

TABLE II

Integer MV Ratio in Final MV (10 frames, QP=28)

Sequence	Integer MV ratio	Sequence	Integer MV ratio
Tempete	84.65%	Bus	76.43%
Silent	81.39%	Stefan	38.26%
Container	80.87%	Football	24.36%

IV. Improved Fast Fractional Pel Motion Estimation Algorithm

A. Inefficient fractional pel search skip algorithm

Based on the analysis of section III, how to predict smooth region before fractional pel search, is the key to skip inefficient fractional pel search. Experiments show that seven ME modes in H.264 have strong correlation. ME result of up layer block can be used to predict movement complexity of the current macro-block, that is, if MV of up layer block (ME mode1 16×16) is at integer pel position, such macro-block is defined as smooth macro-block (SMB). For SMB, fractional pel search of lower layer block can be skipped.

In this paper, we propose an inefficient fractional pel search skip algorithm based on smooth macro-block prediction (SMBP for short). A macro-block is regarded as SMB if MV of mode 1 is at integer position. For SMB, ME modes followed skip fractional pel search; for macro-block which is not SMB, conventional integer and fractional pel search is conducted. Seven modes for MC of H.264 in Fig.2.

Accuracy of SMB prediction is defined as the rate of the final MV located at integer position if conventional fractional pel search is conducted in low layer modes. Statistics of accuracy of SMB prediction, integer MV rate is shown in Table III.

$$\text{Accuracy} = \frac{\text{Num of SMB that final mv is at integer position}}{\text{Num of SMB}} \times 100\% \quad (2)$$

TABLE III

Accuracy of SMB Prediction (10 frames, QP=28)

Sequence	Accuracy	Integer MV rate
Tempete	97.63%	92.67%
Silent	98.21%	91.81%
Container	96.42%	86.36%
Bus	91.43%	88.69%
Stefan	56.54%	65.56%
Football	33.63%	53.70%

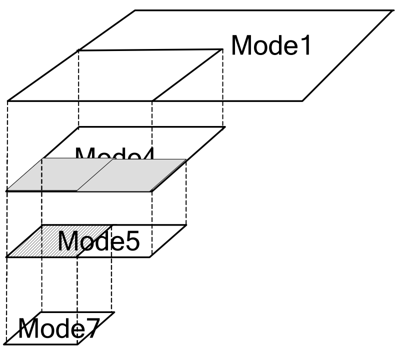
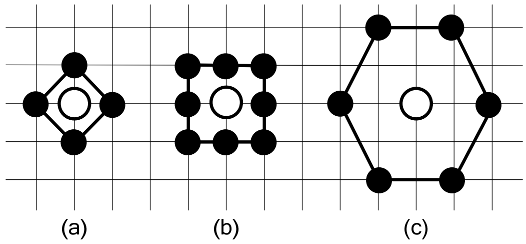


Fig. 2. Seven modes for MC of H.264

The higher the accuracy of SMB prediction, the less the matching error, the deterioration of decoded image quality, and the change of bit rate. As is shown in Table 3, accuracy of SMB prediction is more than 96.42% for sequences of less detail and complex movement.

B. Extended diamond pattern search algorithm based on MVP

As Fig.3 shows, there are three patterns in motion estimation in general: diamond, square and hexagon. Diamond pattern is the simplest one, which is adopted by many codec. Square pattern adds four diagonal positions to diamond pattern, which provides more precise result with additional computation. Hexagon pattern is suitable for situation of large search region, hence is not useful in fractional pel refinement.



(a) Diamond (b) Square (c) Hexagon

Fig. 3. Three patterns in ME

Based on the analysis in section III and above, we proposed an extended diamond pattern search algorithm based on MVP: because of the high matching degree of FMVP and best fractional pel MV, we skip the search center (0, 0) and take FMVP as the initial search center, which is different with CBFPS; we take advantage of high accuracy of square pattern, and add diagonal position to diamond pattern which is called extended diamond search pattern (EDSP for short); we do not perform iteration of diamond pattern, instead, stop fractional pel search in [-2, 2] of FMVP, which skip those fractional pel search out of [-2, 2] that contribute little to the coding efficiency, hence we save computation further.

Fig.4 illustrates the implementation of extended diamond pattern search algorithm based on MVP, the whole algorithm can be described in following steps:

Step 1. Predict the fractional pel motion vector of the current macro-block by neighbor blocks, the predicted motion vector is (Pred_x, Pred_y) and is taken as the search center directly;

Step 2. Cost of (Pred_x, Pred_y) and four diamond positions around are compared. If the MSAD point is located at (Pred_x, Pred_y), stop fractional pel MV search; otherwise go to step 3.

Step 3. As shown in Fig.4, if the best matching point and the second best matching point are at the opposite position, then the best matching point is selected as the final MV; if the best matching point is next to the second best matching point, an additional point in the square pattern will be checked. If MSAD is still in the diamond pattern, the best matching point is selected as the final MV, otherwise go to next step.

Step 4. Choose the point in square pattern in step 3 as the center, check the points around by diamond pattern. Choose the MBD point as the final MV.

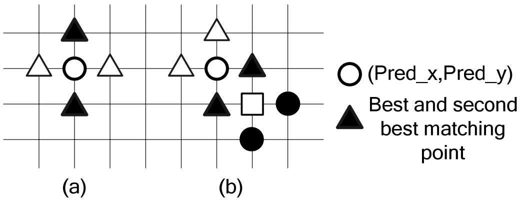


Fig. 4. Extended diamond pattern search based on MVP

V. EXPERIMENTAL RESULTS

To examine the proposed improved fast fractional pel motion estimation algorithm for H.264, we choose six sequences of movement of various complexity. The experiment is proceeded in a PC (CPU: Inter Core2 E6300, 1.86GHz, RAM: 2G, DDR2). Our proposed algorithm is integrated within version 15.0 of the JVT software [5]. The main test conditions of encoder are shown in Table IV.

TABLE IV
MAIN TEST CONDITIONS OF ENCODER

Num of Frames	20
Resolution	CIF: 352×288
GOP Structure	IPPP…
QP	28
Search Range	16
Integer ME Algorithm	UMHexagonS
Symbol Mode	CAVLC

TABLE V
COMPARISON OF ALGORITHMS

Sequence	Algorithm	SP	△ PSNR/dB	△ Time
Tempete	FFPS	16	-	-
	CBFPS	5.4	-0.005	25%
	EDSP	**4.5**	**-0.034**	**32%**
	SMBP+EDSP	**4.5**	**-0.092**	**47%**
Silent	FFPS	16	-	-
	CBFPS	5.3	-0.004	29%
	EDSP	**5.0**	**-0.017**	**30%**
	SMBP+EDSP	**5.0**	**-0.085**	**47%**
Container	FFPS	16	-	-
	CBFPS	5.5	-0.008	28%
	EDSP	**4.6**	**+0.078**	**31%**
	SMBP+EDSP	**4.6**	**+0.020**	**46%**
Bus	FFPS	16	-	-
	CBFPS	5.6	+0.008	20%
	EDSP	**4.3**	**+0.045**	**23%**
	SMBP+EDSP	**4.3**	**+0.015**	**35%**
Stefan	FFPS	16	-	-
	CBFPS	6.9	-0.017	16%
	EDSP	**3.4**	**-0.016**	**25%**
	SMBP+EDSP	**3.4**	**-0.183**	**35%**
Football	FFPS	16	-	-
	CBFPS	8.2	-0.013	9%
	EDSP	**2.1**	**-0.118**	**19%**
	SMBP+EDSP	**2.1**	**-0.189**	**27%**

The proposed EDSP, SMBP+EDSP algorithms are compared against FFPS and CBFPS adopted by H.264 JM software by: (1) Average number of fractional pel search points per macro-block (SP) which reflects the matching speed; (2) Peak signal noise ratio (PSNR) which reflects the accuracy of prediction. (3) △computation time (time saved of integer and fractional pel ME) which reflects the time consumed in ME module of encoder on the whole.

Table V shows that SP of each macro-block is 4.0 on average of EDSP algorithm which saved 75% compared with FFPS with PSNR loss less than 0.12dB. Compared with EDSP, SMBP+EDSP saved 16% computation for sequences of less detail and complex movement; 11% for sequences of medium detail and complex movement; 8% for sequences of high detail and complex movement. Compared with optimal algorithm FFPS, SMBP+EDSP saved 39.5% of computation time on average with PSNR loss less than 0.189dB. Compared with CBFPS, SMBP+EDSP saved 18.3% of computation time on average with PSNR loss less than 0.176dB.

VI. CONCLUSION

This paper first proposed an inefficient fractional pel search skip algorithm based on smooth macro-block prediction. SMBP algorithm takes full advantage of correlation of seven prediction modes in H.264, which predicts SMB according to MV of mode 1. For SMB, other six modes skip fractional pel search. Experiment shows that the proposed algorithm can reduce fractional pel MV search by 28.70%~56.00% with almost the same decoded image quality. The SMBP algorithm is independent, which can be integrated with extended diamond pattern search algorithm based on MVP and reduce computation of fractional pel ME further. Compared with optimal algorithm FFPS, SMBP+EDSP saved 39.5% of computation time on average with PSNR loss less than 0.189dB, Especially, for sequences of less detail and complex movement, SMBP+EDSP saved 47% of computation time with the same or better PSNR.

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